



Digital Twin Manager

Last updated: Wednesday, August 20, 2025

This content applies to Cumulocity and to all subsequent releases.

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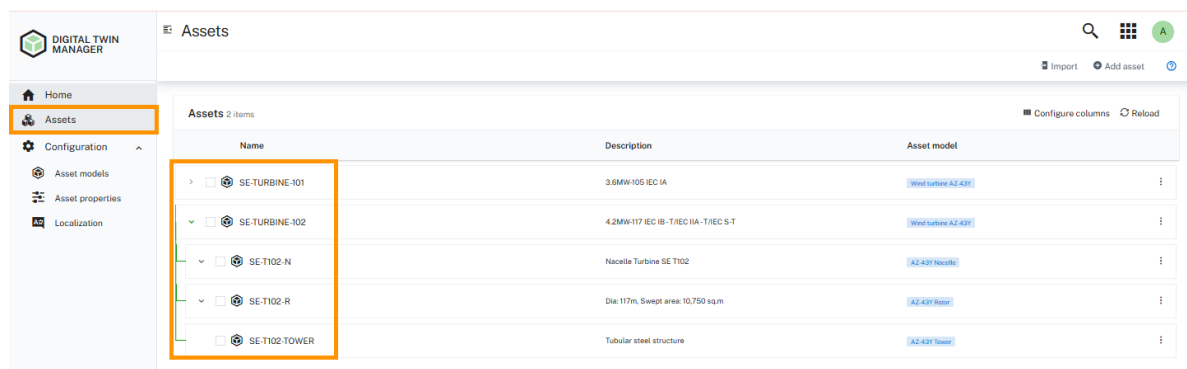
INTRODUCTION

ASSETS

In the DTM application assets are the digital representation of physical assets. An asset in the DTM application can contain subassets and devices. You also have the opportunity to define properties or parameters of an asset through [asset properties](#).

To work with assets in the DTM application, navigate to the **Assets** page using the navigator menu on the left. When you first start using the DTM application, no assets are displayed in the **Assets** page by default.

If you already created assets, you find them listed in a hierarchical view on this page.



The screenshot shows the 'Assets' page in the Digital Twin Manager application. The left sidebar contains a navigation menu with 'Assets' highlighted. The main area displays a table of assets with columns for Name, Description, and Asset model. The assets are organized in a hierarchical tree structure. An orange box highlights the following assets:

Name	Description	Asset model
SE-TURBINE-101	3.6MW-105 IEC IA	Wind turbine AZ-43T
SE-TURBINE-102	4.2MW-117 IEC IB-T/IEC IIA-T/IEC S-T	Wind turbine AZ-43T
SE-T102-N	Nacelle Turbine SE T102	AZ-43T Nacelle
SE-T102-R	Dia: 117m, Swept area: 10,750 sqm	AZ-43T Rotor
SE-T102-TOWER	Tubular steel structure	AZ-43T Tower

REQUIREMENTS

ROLES & PERMISSIONS

- To view all assets: READ permission for permission type "Inventory"
- To add/update/delete assets: CREATE/ADMIN permission for permission type "Inventory"
- To view specific assets: READ permissions for "Inventory" in the inventory roles
- To manage or delete specific assets: READ and CHANGE permissions for "Inventory" in the inventory roles

Note that global inventory permissions override inventory role permissions. By default, the user has full access to assets created by them regardless of permissions granted to them. See [Managing permissions](#) for

further information.

ASSET HIERARCHY

An asset hierarchy is a structure that organizes multiple assets and devices into a hierarchy, allowing for the visualization and management of complex systems.

The asset hierarchy is displayed in the **Assets** page. An asset hierarchy consists of a root asset and all its subsequent subassets. Click **Add asset** to create the entire hierarchy of an asset in the **New asset** page. For details, see [To create an asset](#).

Example:

If you create the root asset “Wind turbine” with the subasset “Rotor”, which has a subasset “Blade” then the hierarchy would be as follows:

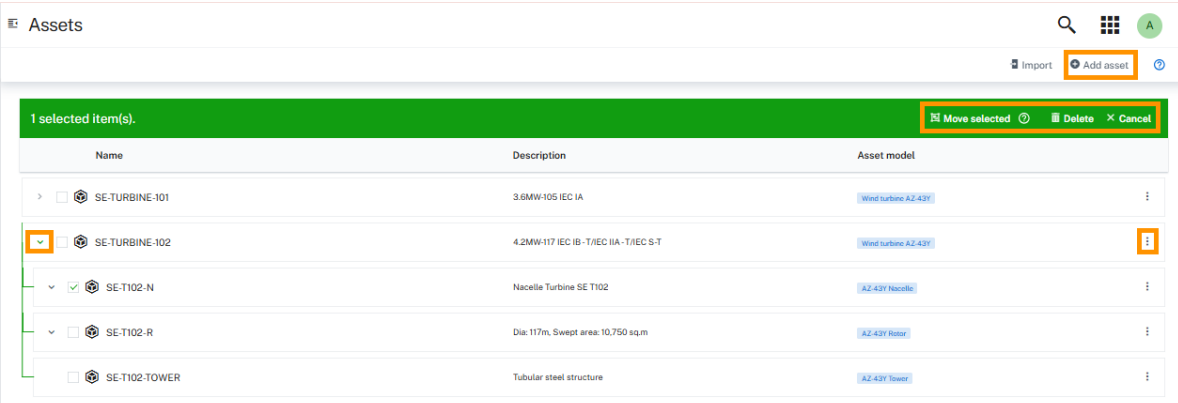
Wind turbine > Rotor > Blade

Initially, all the asset hierarchies are collapsed and it lists only the root assets. The list also shows certain asset details, for example, asset model and description. Click the expand icon  to the left of each asset to view the next level of subassets. Note that the devices assigned to an asset is not shown in the asset hierarchy.

Click the asset name to view the asset details and the devices assigned to it.

To edit or delete an asset from the **Assets** page, click the menu icon  at the right of the row and click either **Edit** or **Delete**. To delete multiple assets at once, select one or more assets and click **Delete** in the top banner.

Select one or more assets of same type to relocate from its current hierarchy to another. For more information on how to move assets between hierarchies, see [To move assets](#).



Name	Description	Asset model
>  SE-TURBINE-101	3.6MW-105 IEC IA	Wind turbine AZ-43Y
 SE-TURBINE-102	4.2MW-117 IEC IB - T/IEC IIA - T/IEC S-T	Wind turbine AZ-43Y
 SE-T102-N	Nacelle Turbine SE T102	AZ-43Y Nacelle
 SE-T102-R	Dia: 117m, Swept area: 10,750 sq.m	AZ-43Y Rotor
 SE-T102-TOWER	Tubular steel structure	AZ-43Y Tower

CREATING ASSETS VIA THE UI

You have two options for creating assets in the DTM application. The following section describes how to create an asset via the UI. Alternatively, see [Creating assets via bulk import](#) on how to import assets in bulk into the DTM application.

TO CREATE AN ASSET

To add an asset via the UI:

1. Click **Add asset** in the top right corner of the **Assets** page.
2. In the **New asset** page, select the desired asset model from the **Choose asset model** dropdown menu on the top left.

INFO

The asset models here are root asset models. They are labeled "START NODE" in the hierarchy. Root asset models are asset models which are at the top of hierarchy and don't have any parent asset model above them.

3. The asset hierarchy is shown in the left section and the respective dialog window is shown on the right. Fill out the fields labeled "required".
4. Optionally, add a description or [assign devices](#) to your root asset.
5. To add more than one root asset, click **Add**.
6. Under **Asset properties**, fill in the required information. Here you find all [asset properties](#) assigned to the asset model that you create.

For an asset with a **Location** property, click **Choose on Map** in the section to set the values for the latitude and longitude using map view. Click the full screen icon  at the top right corner of the map to view it in full screen.

The marker is positioned at the default value set in the **Location** property. Click or drag the marker to the preferred position to set the value for both latitude and longitude. If you do not enter a latitude or longitude value, you will not see a marker on the map.

INFO

The values of latitude and longitude are automatically updated whenever a new location is selected on the map and vice-versa. The altitude value is not represented on the map.

The asset properties are defined when [creating the asset model](#). When creating an asset you must provide the values for all asset properties.

For complex properties, the keys are ordered as specified in the property definition.

7. Click **Next** to reach the next hierarchy level of your **asset hierarchy** and fill out the fields as necessary for your asset. Repeat for all asset hierarchy levels.
8. The **Confirmation** step displays an overview of your asset and asset hierarchy levels. Click **Create** to create your asset. The asset is now listed in the **Assets** page.

As you define each hierarchy level of your new asset, you see a green check mark on each asset hierarchy level in the section on the left.

INFO

If there are validation errors on the page, they are indicated by red highlights. Correct the required information to continue. If you are leaving the page abruptly, a confirmation dialogue box is shown.

The screenshot displays the 'Wind turbine AZ-43Y' asset hierarchy on the left and the 'Wind turbine AZ-43Y required' configuration form on the right.

Asset hierarchy (left): A tree view showing the hierarchy levels. The root node is 'Wind turbine AZ-43Y' (START NODE). Below it are 10 levels, each with a green checkmark indicating completion. The levels are: 1. Wind turbine AZ-43Y (START NODE), 2. AZ-43Y Tower, 3. AZ-43Y Nacelle, 4. AZ-43Y Low speed shaft, 5. AZ-43Y Gearbox, 6. AZ-43Y Generator, 7. AZ-43Y High speed shaft, 8. AZ-43Y Rotor, 9. AZ-43Y Blade, and 10. Confirmation.

Wind turbine AZ-43Y required (right): A form with the following fields and values:

- Name:** SE-TURBINE-101
- Description:** 3.6MW-105 IEC IA
- ASSET PROPERTIES:**
 - Model name:** AZ-43Y
 - Installation date:** 2021-07-09
- Operating data:**
 - Rated power:** 3.560 kW
 - Wind class:** IEC IA
 - Standard operating temperature:** -20 C to +45 C
 - Cut-out wind speed:** 25 m/s

At the bottom of the form are two buttons: 'Cancel' and 'Next'.

Example:

If you are creating an asset hierarchy for the asset model "Wind turbine AZ-43Y", select the root asset model "Wind turbine AZ-43Y" in the **Choose asset model** dropdown menu. Starting with the root asset level, you see the dialog window "Wind turbine AZ-43Y" on the right. Enter the name of the wind turbine in the field **Name**, for example, "SE-TURBINE-101". In the field **Description** you can enter a brief description of this asset, for example, "Wind turbine with rated power of 3.6MW". However, this field is optional and can be left empty. To add more assets to this level, click **Add new** at the bottom. Repeat until all desired assets on this level have been filled out.

Click **Next** to see the next asset level, for example, "Rotor". Fill out all required fields. You have the option to add more assets on this level. Click **Next** to continue until all assets are created.

TO ASSIGN DEVICES TO AN ASSET

When you create a new asset you have the option of assigning one or more devices to this asset.

1. Click **Assign devices** in the **New asset** page.
2. The resulting dialog window lists all devices registered for the tenant. Select one or multiple devices and click **Assign**.
3. The dialog closes and you can continue creating the asset.

Assign devices

2 selected items. ✕ Cancel

Status	Name	Model	Serial number	Group	Registration date	System ID	IMEI	Alarms ⓘ
☐	LWM2M t1458741 connector				29 Nov 2024, 18:41:28	46295200		
☒	PitchCtrl-SE-T101-R				2 Dec 2024, 17:53:58	71298207		
☒	Tmeter-SE-T101-R	AP14875	N102548K9908		2 Dec 2024, 17:54:15	85297205		
☐	Tmeter-SE-T102-R	AZ 1419	L10150009		2 Dec 2024, 18:04:09	13297206		

Cancel Assign

To add a new device, add it through the Device Management application. Refer to [Registering devices](#) for further information.

ⓘ INFO

Only select devices which are part of the current asset. If a device belongs to a subasset, then select it when you create the subasset.

To filter and select devices

You can view, search or filter devices easily with the following options:

1. Columns in the grid specify the device details for each device. Click **Configure columns** at the top right to show or hide columns.
2. Click **Reload** to reload the page and display the latest list of devices present in the Cumulocity tenant.
3. Click **Sort** on the applicable columns to view the device data in either ascending or descending order.

- To filter devices based on text, use the **Filter** option in the applicable columns. If you apply a filter, you see a notification at the top.
- To clear the filters, click **Clear all filters**. To see the sort icon $\updownarrow$ and the filter icon ∇ hover over each column.

TO ADD MULTIPLE INSTANCES OF AN ASSET

At each hierarchy level, you can create multiple assets from a single asset model.

Example:

For a wind turbine rotor with three blades, you first create the asset hierarchy level for “Rotor”. Then use the asset model “Blade” to add the blades.

- Click **Add** at the bottom to add more blade assets.

The screenshot displays the 'Asset hierarchy' panel on the left and the 'AZ-43Y Blade' configuration form on the right.

Asset hierarchy panel:

- Search bar: Wind turbine AZ-43Y
- Tree structure:
 - START NODE: Wind turbine AZ-43Y (checked)
 - AZ-43Y Tower (checked)
 - AZ-43Y Nacelle (checked)
 - AZ-43Y Low speed shaft (checked)
 - AZ-43Y Gearbox (checked)
 - AZ-43Y Generator (checked)
 - AZ-43Y High speed shaft (checked)
 - AZ-43Y Rotor (checked)
 - AZ-43Y Blade (checked)
 - 10 Confirmation (checked)

AZ-43Y Blade configuration form:

- Name:** SE-T101-B
- Description:** length: 52m
- Parent:** SE-T101-R/SE-TURBINE-101
- ASSET PROPERTIES:**
 - Blade length:** 52
- ASSIGNED DEVICES:**
 - No devices assigned.
 - Assign devices to this asset to monitor measurements, events and alarms.
 - [Assign devices](#)
- Buttons:** Add (highlighted with an orange box), Cancel, Previous, Next

- To delete an added asset template, click the delete icon at the top right.
- Modify the assets by navigating the asset pages using the **Previous** and **Next** buttons. You can track your progress via the green check marks in the asset hierarchy on the left.
- When done, the **Confirmation** page shows the asset hierarchy. Click **Create** to create the asset hierarchy.

Asset hierarchy

Wind turbine AZ-43Y

- ✓ START NODE
Wind turbine AZ-43Y
- ✓ AZ-43Y Tower
- ✓ AZ-43Y Nacelle
- ✓ AZ-43Y Low speed shaft
- ✓ AZ-43Y Gearbox
- ✓ AZ-43Y Generator
- ✓ AZ-43Y High speed shaft
- ✓ AZ-43Y Rotor
- ✓ AZ-43Y Blade
- ✓ Confirmation

Confirmation

You will create "9" asset(s) with the following hierarchy:

- SE-TURBINE-101
 - SE-T101-TOWER
 - SE-T101-N
 - SE-T101-LSS
 - SE-T101-GB
 - SE-T101-GEN
 - SE-T101-HSS
 - SE-T101-R
 - SE-T101-B

Cancel

Previous

Create

On successful asset hierarchy creation, you see a pop-up notification in the top right corner. You can view the newly created asset hierarchy in the **Assets** page. Also see [viewing assets](#).

In case of a bulk operation failure, the entire operation is rolled back and the user will be navigated to main assets page to cleanup the entities created and to try the operation again.

TO VIEW ASSET DETAILS

Select an asset in the **Assets** page to view the asset details in **Subassets** tab. It shows all the subassets, child devices and asset properties.

INFO

The key value pairs for complex properties in the **Subassets** page are displayed in the order which has been specified in the **Order** field of property definition in the **Asset properties** page.

In the top right corner of the **Subassets** tab, you can [assign devices](#).

The screenshot shows the 'SE-TURBINE-101' page with the 'Subassets' tab selected. The top right corner has buttons for 'Assign devices' and 'Add asset'. The main area displays a table of subassets:

Type	Name	Model	Alarms
	SE-T101-N		
	SE-T101-R		
	SE-T101-TOWER		

On the right, the 'Wind turbine AZ-43Y properties' panel shows:

- Model name:** AZ-43Y
- Installation date:** Friday, 9 July 2021
- Operating data:**
 - Rated power: 3.560 kW
 - Wind class: IEC IA
 - Standard operating temperature: -20 C to +45 C
 - Cut-out wind speed: 25 m/s
 - Cut-in wind speed: 3 m/s
- Platform capacity:** 3.5

You can view the hierarchy of the asset in the **Asset tree** tab. Initially the asset hierarchy is collapsed. Click the expand icon to view the next level of subassets.

Click **Add asset** to add more subassets or click **Import assets** to add [assets in bulk](#).

The screenshot shows the 'SE-TURBINE-101' page with the 'Asset tree' tab selected. The top right corner has buttons for 'Import' and 'Add asset'. The main area displays a tree view of assets:

Name	Description	Asset model
SE-TURBINE-101	3.6MW-105 IEC IA	Wind turbine AZ-43Y
SE-T101-N	Nacelle turbine SE T101	AZ-43Y Nacelle
SE-T101-R	Dia: 105m, Swept area: 8650 sq.m	AZ-43Y Rotor
SE-T101-TOWER	Tubular steel structure	AZ-43Y Tower

To add subassets

To add subassets to an asset open the **Subassets** or **Asset tree** tab.

1. Click **Add asset** to load the **Asset hierarchy** for the selected asset.
2. The root asset is displayed with the label "parent node". The asset hierarchy displayed below, is a subset of the root asset hierarchy.

New asset

Asset hierarchy

- ✓ **SE-TURBINE-101** (Parent Node)
- ✓ AZ-43Y Tower
- 3 AZ-43Y Nacelle
- 4 AZ-43Y Low speed shaft
- 5 AZ-43Y Gearbox
- 6 AZ-43Y Generator
- 7 AZ-43Y High speed shaft
- 8 AZ-43Y Rotor
- 9 AZ-43Y Blade
- 10 ☒ Confirmation

AZ-43Y Tower

Name

Description

Parent

SE-TURBINE-101 ✕ ▼

ASSIGNED DEVICES

→ No devices assigned.
← Assign devices to this asset to monitor measurements, events and alarms.

⚙️ Assign devices

➕ Add

Cancel Next

Example:

If “Wind turbine” is a root level asset, then “Rotor” is a subasset of “Wind turbine” and “Blade” is a subasset of “Rotor”. If you navigate to the **Subassets** or **Asset tree** page of “Wind turbine” and click **Add asset**, the asset hierarchy loads for “Rotor” and “Blade”.

If you navigate to the existing “Rotor” asset and want to add a subasset, you can only add subassets for “Blade”.

For details on how to add assets in the asset hierarchy, see [To create an asset](#).

i INFO

If you are at the end of the hierarchy, you see no further option to create subassets.

CREATING ASSETS VIA BULK IMPORT

To import the entire asset hierarchy with all assets at once, use the bulk import feature. For each [asset](#)

[model](#), a CSV template is provided. Fill in the required details in this template and upload the file to create the assets in bulk for the selected asset model.

TO USE THE BULK IMPORT FEATURE

To use the Bulk import feature, you must enable the **Digital Twin** permission for the user's role, following these steps:

1. Navigate to the [Administration application](#) via the application switcher in the top right corner.
2. Under **Quick links** click **Roles**.
3. In the **Global roles** tab, select the role to which permissions are to be provided.
4. Select the "Admin" permission for "Digital Twin" in the "Permissions" section.
5. Enable access to the DTM application by selecting the following checkboxes:
 - Under **Application access**, select "Dtm-ms".
 - Under **Custom applications**, select "Digital Twin Manager".
6. Click **Save**.

The screenshot displays the 'Global roles' configuration page for a 'Business User' role. The 'Permissions' table is as follows:

TYPE	READ	ADMIN	CREATE	UPDATE
CEP management	☐	☐	☐	☐
Data broker	☒	☐	☐	☐
Device control	☒	☒	☐	☐
Digital twin	☐	☒	☐	☐
Email	☐	☐	☐	☐
Eplapps	☐	☐	☐	☐
Events	☒	☒	☐	☐
Identity	☒	☒	☐	☐
Inventory	☒	☒	☐	☐

The 'Application access' sidebar on the right shows the following checked items:

- ☒ Administration administration
- ☒ Cockpit cockpit
- ☒ Device Management device management
- ☒ Feature-branding feature branding
- ☒ Feature-cep-custom-rules feature-cep-custom-rules
- ☒ Feature-microservice-hosting feature-microservice-hosting
- ☒ Public-options public-options
- ☒ Ui-assets-t7690 ui-assets-t7690
- ☒ Digital Twin Manager digital-twin-manager

Once the permissions are assigned, the bulk import feature can be used.

i INFO

If the permissions are not assigned, the CSV template will not be downloaded correctly.

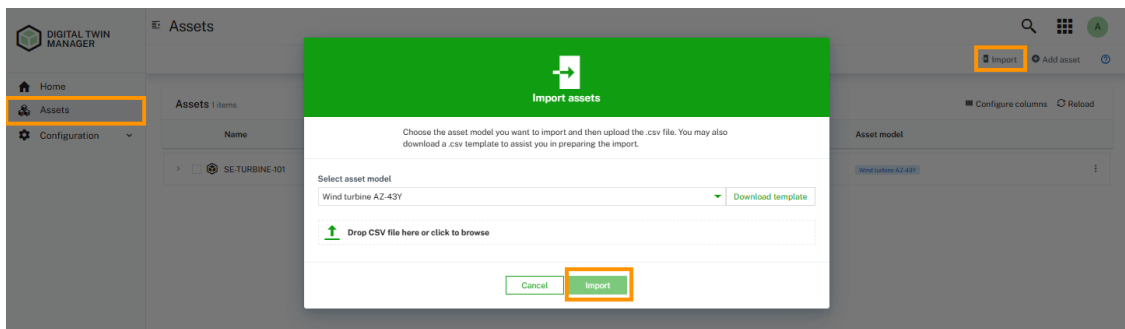
To initiate a bulk import from the **Assets** page, you must either have CREATE or ADMIN permission for the permission type "Inventory".

By default, you have full access to assets created by you regardless of permissions granted. To perform a partial import on an asset created by other users, you will need READ and ADMIN permissions for permission type "Inventory" or alternatively, READ and CHANGE permissions for "Inventory" in the inventory roles for the specific asset.

TO DOWNLOAD A CSV TEMPLATE FOR BULK IMPORT

Follow the steps below:

1. Navigate to the **Assets** page and click **Import**.
2. The **Import** dialog has a dropdown **Select asset model**. If you have added the asset models, all the root asset models are listed in the dropdown.
3. Select the asset model for which you want to create the asset hierarchy.



4. On selection, the following options become available:
 - **Drop CSV file here or click to browse** - upload the CSV template as a file, for importing assets in bulk.
 - **Download Template** - download the CSV template for the selected asset model.
5. Click **Download Template** to download the CSV template.

CSV TEMPLATE PARAMETERS

The CSV template has the following fields:

Field	Description	Mandatory / Optional
AssetModel / DeviceType	Enter the key of the asset model.	Mandatory
AssetName	Provide the name of the asset you want to create.	Mandatory

Field	Description	Mandatory / Optional
Path	Path denotes the location of the asset within the asset hierarchy. For a root asset, it remains blank. For a subasset, provide a path up to the root asset. Example: For an asset called "SE-T101-Blade 1" in the hierarchy "Wind turbine > Rotor > Blade" the path value is "SE-Turbine-101/SE-T101-Rotor", where "SE-Turbine-101" and "SE-T101-Rotor" each represent the respective asset models. In case of partial import, the path is defined in relation to the asset from which the file is imported. For the immediate children of the asset, the path remains blank. Example: In the hierarchy, "Wind turbine > Rotor > Blade", to import a new Rotor, "SE-T102-Rotor," and a Blade, "SE-T101-Blade2," under the existing Turbine "SE-Turbine-101", the path value for the Rotor remains blank, and the path value for the Blade is "SE-T102-Rotor."	Mandatory (for all subassets) Mandatory (for subassets except immediate children in the context of partial import)

Field	Description	Mandatory / Optional
Device ID / External ID	If the asset being created has a device associated with it, then provide the Device ID of the device here. Note that you can use a list of semi-colon separated Device IDs or External IDs to associate multiple devices to the asset.	Optional
Description	briefly describes the asset being created.	Optional
Asset properties for the asset model	6th column onwards (in the CSV template), all the asset properties for the root asset models and all its subsequent child asset models are listed. Note that the Asset property type is also mentioned as a label, for better understanding.	Mandatory (If the property is set as required in the asset model. For details, see To add an asset property to an asset model).

AssetModel/DeviceType	AssetName	Path	Device ID/External ID	Description	Wind turbine AZ-43Y.Model name(string)	Wind turbine AZ-43Y.Installation date(date)(yyyy-mm-dd)	Wind turbine AZ-43Y.Operating data.Cut-in wind speed(string)	Wind turbine AZ-43Y.Operating data.Cut-out wind speed(string)
wind_turbine_az-43y	SE-TURBINE-102		1218971	4.2MW-117 IEC IB - T/IEC IIA - T/IEC S-T	AZ-43Y-1A	09-07-2022	4200kW	
az-43y_tower	SE-T102-TOWER	SE-TURBINE-102		Tubular steel structure				
az-43y_nacelle	SE-T102-N	SE-TURBINE-102		Nacelle Turbine SE T102				
az-43y_low_speed_shaft	SE-T102-LSS	SE-TURBINE-102/SE-T102-N		Low speed shaft				
az-43y_gearbox	SE-T102-GB	SE-TURBINE-102/SE-T102-N		Hybrid - Planetary and helical				
az-43y_generator	SE-T102-GEN	SE-TURBINE-102/SE-T102-N		Rated power 3560 kW				
az-43y_high_speed_shaft	SE-T102-HSS	SE-TURBINE-102/SE-T102-N		High speed shaft				
az-43y_rotor	SE-T102-R	SE-TURBINE-102		Dia: 117m, Swept area: 10,750 sq.m				
az-43y_blade	SE-T102-B1	SE-TURBINE-102/SE-T102-R						
az-43y_blade	SE-T102-B2	SE-TURBINE-102/SE-T102-R						
az-43y_blade	SE-T102-B3	SE-TURBINE-102/SE-T102-R						

INFO

Use only comma (,) as a separator when entering values in the CSV template.

Modify the excel settings to provide the date in a YYYY-MM-DD format. This prevents auto-correction of the date on input.

To provide a file input as an asset property value, the file must first be uploaded to a Cumulocity tenant using the Cumulocity API.

Refer to the [Binaries API](#) in the Cumulocity OpenAPI Specification for details on how to upload a file to Cumulocity.

The binary ID in the API response must be provided as input for the asset property field with type “file” in the CSV template.

INFO

During the bulk import, the file size validation is skipped as the file is already uploaded to the tenant.

If the type is “boolean”, the input field must be either “true” or “false”. If the type is “enumeration”, then the input field must be in the list of values specified during creation of the asset property. If the type is “text” or “number” and custom criteria were provided during asset property creation, then the input value in CSV template must fulfill all the asset property criteria.

When creating a property, you have the option to assign a default value. If a property’s value is left empty in the CSV file during the bulk import, its value is set to default. To override this behavior, enter ‘null’ as the value in the CSV file. This does not set the value and it is seen as undefined when viewing the asset.

Fill in details for all the assets which must be created as part of the asset hierarchy.

Start with the asset details for root asset model. Then enter the asset details for child models under the root asset. Next, enter the details for the next level in the hierarchy of subassets and repeat until the details for all the assets are entered in the CSV template. Save the CSV template in your system.

INFO

It is recommended to import a maximum of 5000 assets at a time for a better user experience.

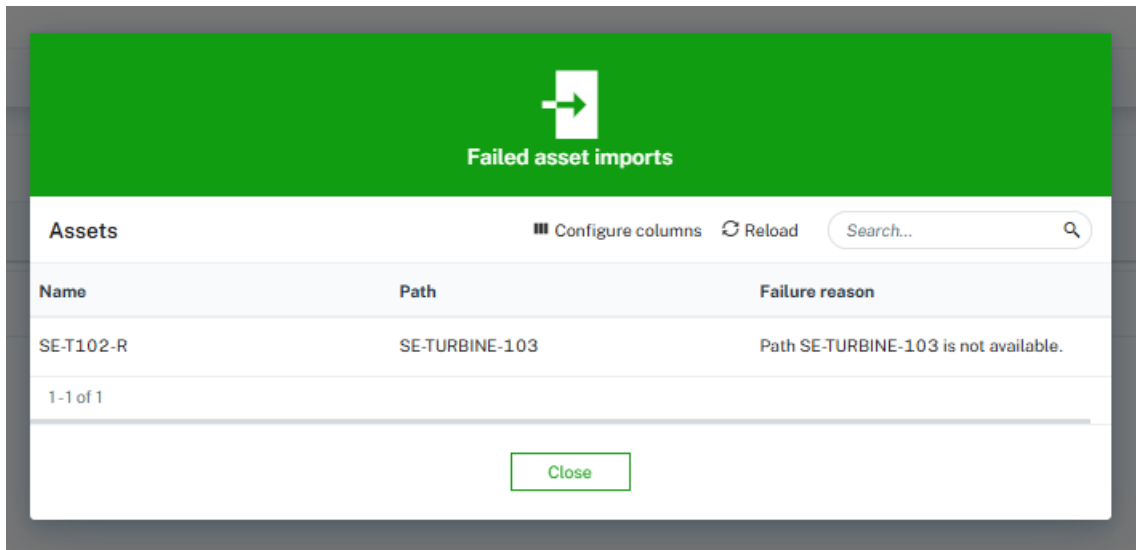
TO UPLOAD A CSV TEMPLATE

To upload the CSV template follow the steps below:

1. Click **Import** in the **Assets** page.
2. Select the asset model in the **Import** dialog.
3. Upload the previously filled and saved CSV template in the **Drop file here** section and click **Import**.
4. If all the inputs are provided correctly, the bulk import is successful, resulting in a success notification. The asset hierarchy and its assets are created successfully.
5. View the newly created assets in the **Assets** page.

If the inputs are not provided in the correct format, the bulk import fails with validation errors, resulting in a **Failed asset imports** notification, showing the issue types, for example, an invalid template or an

incorrect asset hierarchy. No assets are created.



If the bulk import fails, review the reported issues and try again.

PARTIAL IMPORT

If the asset hierarchy already exists, and you want to import more assets, it can be achieved by a partial import.

To import subassets or the subasset hierarchy:

1. Navigate to the **Asset tree** page of the subasset and download the template using **Import**.
2. Fill in the details for the subasset hierarchy in the CSV template. See [CSV template parameters](#) for more details.

INFO

From any hierarchy level, the template will be downloaded only for its child hierarchy. Assets must only be imported for the child hierarchy.

3. Click **Import**.
4. Upload the CSV template in the **Drop file here** section in the **Import** dialog and click **Import**.

If all details are mentioned correctly in the CSV template, the subasset hierarchy is created successfully.

An unsuccessful partial import will result in a **Failed asset imports** notification, showing the validation issues. Review the reported issues and try again.

VIEWING ASSETS

Once the [asset hierarchy](#) is created by adding assets, they can be viewed in the **Assets** page.

TO VIEW AN ASSET

1. Select an asset from the hierarchy on the **Assets** page.
2. The **Subassets** page contains the asset details, created and last updated time displayed at the top right.
3. Optionally [add subassets](#), or [assign devices](#) from the top bar.
4. The asset name and description along with the icon are displayed at the top. All subassets and devices for this asset will get listed in the **Subassets** section.
5. All the asset properties for this asset will be displayed on the right.
6. In the **Asset tree** tab you see the asset hierarchy.
7. Optionally [add subassets](#), or [import assets](#) using the options on the top right.

If the asset has a location, you can see it on the map under the properties section below its values.

Click the full screen icon  at the top right corner of the map to view it in full screen. To see the map, provide a value for latitude and longitude. You cannot modify the marker when viewing the asset. Edit the property in order to change the position of the marker.

INFO

The buttons [Import assets](#), and [Add asset](#) are not visible for the last hierarchical level.

You see a warning message if one or more properties associated with the asset are not present. For complex properties, the keys are ordered as specified in the property definition.

TO MODIFY AN ASSET

Fields are marked editable by an edit icon . To modify any of the asset details, click the edit icon , enter new details and save.

For an asset with a location, click or drag the marker to the preferred position to select the value for latitude and longitude on the map. Click the full screen icon  at the top right corner of the map to enter full screen mode. To see the map, provide a value for latitude and longitude.

TO DELETE AN ASSET

To delete a subasset or device in the hierarchy:

1. Click the delete icon  next to it.

! INFO

The delete icon appears on hover of each row under the subassets section.

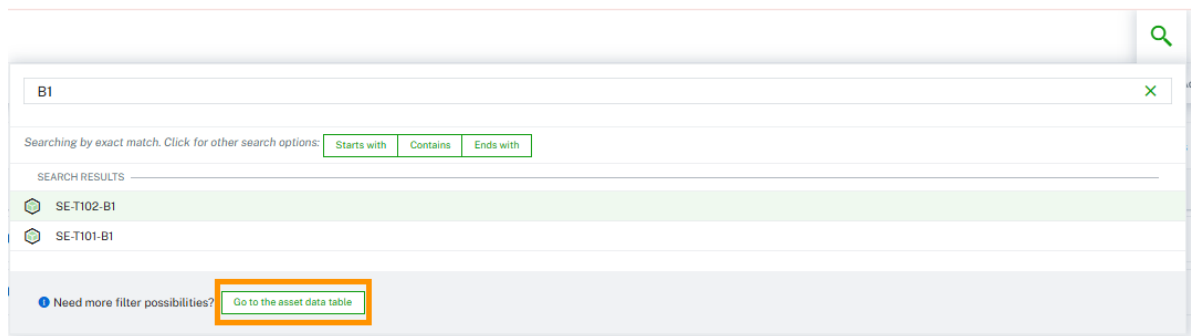
In the following dialog box, click **Confirm** to continue. Optionally select the checkbox in the dialog box to delete all subassets and devices for the selected asset.

TO SEARCH FOR AN ASSET

You can search for assets through the **Search** button at the right of the top bar. Enter a search term into the textbox at the top of the **Search** window to see all assets matching the search criteria in the section **Search results**.

To see more details click **Go to the asset data table** at the bottom. This will show the entire search results in a table format.

The **Search** page only shows a limited number of matches. In case of more matches, switch to the asset data table to see the complete results.



! IMPORTANT

The search results include all assets containing the search term in any property (name, model or any fragment), that is, the search results do not only include assets matching the search criteria with their names.

The DTM application extends the full text search capability of Cumulocity for searching assets. See [Search and filter functionality](#) to know more about available search options.

TO MOVE ASSETS

Use the assets move feature to relocate assets within hierarchies. To move one or more assets, follow the steps below:

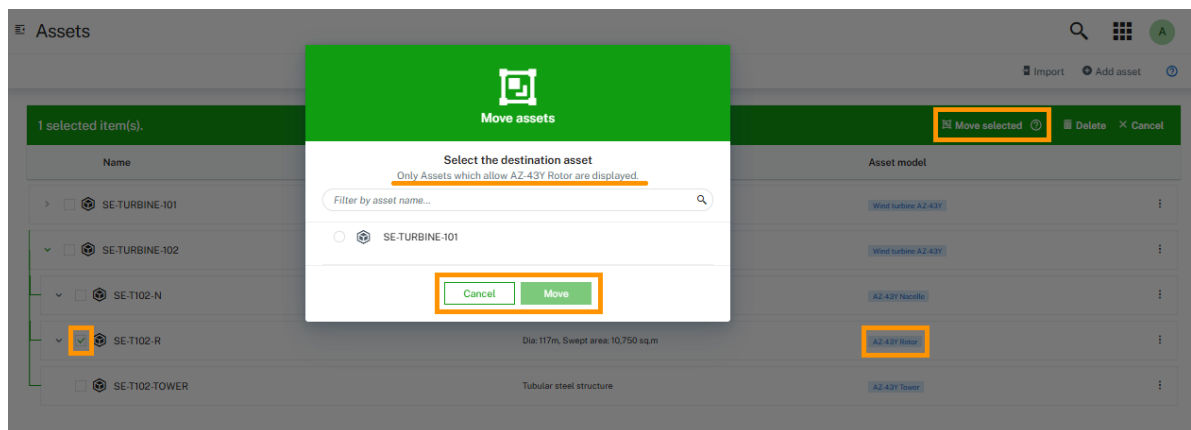
1. Select one or more assets of same type in the **Assets** page to move.
2. Click **Move selected** in the top banner.
3. In the resulting dialog, you see a list of assets which allow selected assets as its children. Click the radio button to the left of each asset to select it.
4. Click **Move** to complete the relocation.

INFO

Only a maximum of 10 assets can be moved at a time.

The **Move selected** option in the banner is disabled in the following scenarios:

- One or more root assets are selected.
- Assets of different asset models are selected. Only assets of same asset model can be moved.
- The selected asset's asset model is no longer a child of its parent asset model.
- One or more assets with the same name are selected.



ASSET MODEL FOR GROUPS

The default group asset model allows any group to be treated as an asset, providing a unified, flexible, and consistent approach to managing assets.

To create an asset from the default group asset model in the **New asset** page, select the **Group (Default)** option from the **Choose asset model** dropdown menu on the top left. For information on how to create an asset, refer to [To create an asset](#).

Characteristics of a group asset

Since the default **Group** asset model does not enforce hierarchy, any group asset can be added to another group asset. However, only root assets created using models other than the default **Group** asset model are allowed under a group asset.

Any asset added to a group asset can be moved out of it using the **Move to the root** option in the **Move assets** dialog.

INFO

- Currently, the [bulk import](#) feature does not support creating groups in bulk.
- Groups created in the Cumulocity Cockpit application are not listed in the Digital Twin Manager application.



ASSET MODELS

ASSET PROPERTIES

Asset properties are the parameters that define an [asset model](#). One or more asset properties are used to define an asset model. Asset properties are created in the **Asset properties** page.

When you open the DTM application, default properties are readily available for use. See [Default properties](#) to view the list of properties available.

🔍 REQUIREMENTS

ROLES & PERMISSIONS

- To add/update/delete/import asset properties: CREATE/ADMIN permission for permission type "Inventory"

TO CREATE AN ASSET PROPERTY

When you navigate to the **Asset properties** page following **Configuration > Asset properties** for the first time, there will be no asset properties present in the system except the default properties. See [Default properties](#) to view the list of properties available.

If you use a tenant which already has asset properties defined, you find the list of defined asset properties on the left. You see the selected asset property definition on the right.

To create an asset property, click **Add asset property** on the bottom left. Enter the required data in the form (see description below) and click **Save**.

Use the Export asset properties and Import asset properties feature to transfer asset properties between tenants. This is useful for replicating the asset properties in multiple tenants. By exporting and importing properties, you can ensure consistency and reduce the time and effort required for manual configuration. See [To export asset properties](#) and [To import asset properties](#) on how to export or import asset properties.

The screenshot shows the 'Asset properties' configuration interface. On the left, the 'Configuration' menu item is selected. The main panel displays a search bar and a list of properties. The 'Add new property' form is active, showing the following fields:

- Label:** Blade length
- Key:** blade_length
- Description:** Length of the turbine blade in meters
- Complex property:** Toggled off
- Type:** Number
- Default value:** 52
- Minimum:** 40
- Maximum:** 110

An 'Add asset property' button is located at the bottom left of the main panel.

Asset properties have the following parameters:

Field	Description	Mandatory / Optional
Label	Refers to the name of the asset property. The asset property is further addressed using the information provided here.	Mandatory
Key	Used to uniquely identify and store the asset property in the DTM application. Note that the key is automatically generated based on the label. You can modify the automatically generated key to suit to your needs only during asset property creation. You cannot edit the key after the asset property is created.	Mandatory
Description	Provides a brief description of the asset property.	Optional
Complex property toggle	Used to create a complex asset property, that contains one or multiple key-value pairs. By default, the toggle is switched off.	Optional

Field	Description	Mandatory / Optional
Type	Defines the type of the asset property. It can be any of the following categories – Text , Number , Date picker , Enumeration , Boolean or File upload .	Mandatory
Default value	Defines the default value for the parameter. You can modify the content when creating an asset using this asset property.	Optional

INFO

The **Default value** field is only shown if the selected **Type** is either “Text” or “Number”.

Once you have defined all necessary asset properties for your project, you must define the [asset models](#).

ASSET PROPERTY TYPES

To further define the asset property you can select one of the six following options listed under **Type**:

Text

Select **Text** if the asset property value is a string, for example, a wind turbine manufacturer name. Enter a valid text.

If you select **Text**, the following fields appear:

- **Min Length:**

Enter the minimum length of the text that must be provided for this asset property during the asset creation. Any text below the minimum length is not accepted.

- **Max Length:**

Enter the maximum length of the text that must be provided for this asset property during the asset creation. Any text above the maximum length is not accepted.

- **RegExp:**

Enter a valid regular expression. During the asset creation, you must provide the asset property value, which adheres to the regular expression.

Example:

If the asset property is “Generator code” with the regular expression “^MCGEN[a-zA-Z0-9]*\$”, all generator codes must start with the prefix “MCGEN”.

If you do not select any of the restrictions above, it is not applicable for the asset property during asset creation.

Default value

Assigns a default value to the asset property. If this field is left empty in the asset property and marked as required in the asset model, you must enter a value during the asset creation.

INFO

The default value must adhere to all validations provided beforehand. If you selected a property of the type “Text”, you can set the **Min length**, **Max length** and/or **RegExp** and the default value must fulfill all set validations.

Example:

If the asset property is “Generator code” with a **Min length** of 8, a **Max length** of 20 and the **RegExp** defined as “^MCGEN[a-zA-Z0-9]*\$”, the default value must fulfill all of these three criteria. If it does not, this results in a validation error. The respective field will be highlighted to modify its content to fit the criteria.

Number

For this type you must provide a numeric value for the asset creation, for example, the height of a wind turbine tower.

The type **Number** contains the following fields for additional information:

- **Minimum:**

If selected, enter a number in the field **Minimum** on the right. When you create the asset, the asset property value cannot be lower or less than provided in the field.

INFO

If you entered a **default value** beforehand, it must be greater than or equal to the value provided in the field **Minimum**. By default, this box is not selected.

- **Maximum:**

If selected, enter a number in the field on the right. When you create the asset, the asset property value cannot exceed the **Maximum** provided.

INFO

If you entered a **default value** beforehand, it must be lesser than or equal to the **Maximum** entered here. By default, this box is not checked.

- **Default value:** Assigns a default value to the asset property. If this field is left empty in the asset property and marked as required in the asset model, you must enter a value during the asset creation.

INFO

If the **Minimum** and/or **Maximum** is checked, the given **Default value** is validated against one or both of these values.

Example:

If the asset property is “Tower height” with a **Minimum** of 80 and a **Maximum** of 110, the **Default value** must be between 80 and 110.

Date Picker

If selected, you must select a date from the calendar displayed in the dialog window during the asset creation.

Example:

If “Installation date” is a property of the “Wind turbine” asset, then you must select the installation date of the wind turbine when creating the asset.

Enumeration

Select **Enumeration** to list several values that apply to this asset property.

If selected, a second field appears below. Here, enter all desired information separated by a comma.

During asset creation, these values appear as options in the dropdown menu. Select one of the options to initialize the asset property with that option.

Example:

If you create an asset property for a wind turbine “drivetrain type” and the type must be either Gearbox or Direct-drive, you must provide these options separated by a comma when creating the asset property.

The screenshot shows the 'Asset properties' configuration interface. On the left, there's a list of existing properties: 'Location' (Object) and 'Blade length' (Number). On the right, the 'Add new property' form is visible. It includes a 'Label' field with 'Divetrain type', a 'Key' field with 'divetrain_type', a 'Description' field, a 'Complex property' toggle, and a 'Type' dropdown menu. The 'Type' dropdown is highlighted with an orange box and is currently set to 'Enumeration'. Below it, there's a 'Values (comma separated)' field with the text 'Gearbox, Direct-drive'.

Boolean

If selected, the presence of the asset property can either be true or false during the asset creation.

An asset with this asset property shows a checkbox. Selecting this checkbox sets the asset property value to true.

Example:

If you create an asset property called “Yaw system” to identify wind turbines that rotate towards the wind direction during upwind conditions, the asset property must be created with the type “Boolean”. When you create an asset for a wind turbine with a yaw system present, it is then initialized to “true”. If there is no yaw system present, the asset is initialized to “false”.

File upload

This option allows you to upload a file during the asset creation.

✓ REQUIREMENTS

ROLES & PERMISSIONS

- To add/update files: CREATE/ADMIN permission for permission type “Inventory”

If selected, enter the allowed file types separated by a comma in the **Allowed file types** field. Provide the maximum file size for uploads in the **Max file size in MB** field. When you create an asset with this asset property, you can only upload files that adhere to the allowed file types with their file size not exceeding the set limit.

i INFO

If there is a file size limit set for the tenant, you see the limit along with the field name for **Max file size in MB**. During the bulk import, the file size validation is skipped as the file is already uploaded to the tenant. To prevent security threats, sanitize your files beforehand.

Example:

To upload the schematic diagram for a wind turbine, you name the asset property “Wind turbine schematic” and specify the file type of the schematic file. Then upload the respective file. Once the asset is created, you can view this file for future reference.

TO EDIT AN ASSET PROPERTY

1. Select the desired asset property in the **Asset properties** to see all the details on the right of the screen.
2. Edit any content of the asset property.
3. Click **Save** to save your changes.
4. If this property is used by any asset model, in the resulting confirmation dialog, select **Confirm** to continue to save your changes.

TO SEARCH AND FILTER ASSET PROPERTIES

1. Navigate to the **Asset properties** page.
2. Enter the desired name in the search field on the top left.
3. Click the search icon .

All asset properties matching the search values are displayed on the screen on the left. If the search entry does not match with any asset properties, you see a notification.

TO DELETE AN ASSET PROPERTY

Select the respective asset property in the **Asset properties** and click **Delete** on the bottom right.

In the resulting confirmation dialog, click **Confirm** to continue.

** INFO**

If the asset property you want to delete is part of an existing asset or an asset model, you see an error notification and the asset property is not deleted.

TYPES OF ASSET PROPERTIES

There are two types of asset properties, both fulfill different requirements:

Simple asset property

A simple asset property consists of a single key value pair. This option is set by default when creating new asset properties.

In the **Asset properties** simple asset properties are listed with their type.

Example:

If the asset property is called “Tower height”, you use the key “tower_height” with the Default value of “70” (see image below).

The screenshot shows the 'Asset properties' configuration page. On the left, a list of properties includes 'Location', 'Blade length', and 'Divetrain type'. The 'Blade length' property is highlighted with an orange box, and its type is set to 'Number'. The main area shows the 'Add new property' form. The 'Label' is 'Tower height' and the 'Key' is 'tower_height'. The 'Description' field is empty. The 'Complex property' toggle is turned on, highlighted with an orange box. The 'Type' is set to 'Number' and the 'Default value' is '70'.

Complex asset property

A complex asset property consists of multiple key-value pairs, meaning, it contains a root object followed by one or more key-value pairs. You can select the complex asset property option by switching on the toggle. If you switch on the toggle, you must define each key-value pair as well as providing a type for each pair. You can arrange the properties in the required order by dragging and dropping them. Hover over the row to see the drag icon. Click and drag the drag icon for reordering. On the right side of each key-value pair, you see an expand icon . Click the expand icon  to see a specific property. Note that when you add a new key-value pair, it automatically expands. To hide it, click the collapse icon .

In the **Asset properties** complex asset properties are listed with the label “Object”.

You see the preview of data model on the right side of properties section.

Example:

If the asset property is called “Gearbox specifications” with the key “gearbox_specifications”, then power, speed increasing ratio and weight are the multiple key-value pairs defining the property “Gearbox specifications”. Switch on the **Complex property** toggle (see image below). Enter the required information for all key-value pairs and click **Save**.

Asset properties
Configuration > Asset properties

Filter properties...

- Location (Object)
- Blade length (Number)
- Divetrain type (Enum)
- Nacelle dimension (Object)**

Add new property

Label: Gearbox specifications Key: gearbox_specifications

Description: Brief description on what this property is about

☒ **Complex property** ⓘ

PROPERTIES

- Power
- Speed ratio

Label: Speed ratio Key: speed_ratio

Type: Number x ▼ Default value:

☐ Minimum ☐ Maximum

PREVIEW OF DATA MODEL

```
{
  "gearbox_specifications": {
    "power": "",
    "speed_ratio": null,
    "weight": null
  }
}
```

Add asset property **Cancel** **Save**

TO EXPORT ASSET PROPERTIES

Follow the steps below:

1. On the **Asset properties** page, click **Export**. In the resulting dialog box, you see the list of asset properties along with their description and type.
2. Select one or more desired asset properties and click **Export**.
3. The dialog closes and a JSON file named 'Export-asset-properties.json' downloads.

Use the downloaded file to import the asset properties into a DTM application on another tenant.

INFO

Default properties are not listed under **Export**. See [Default properties](#) to view the list of properties available. You can export a maximum of 500 asset properties at a time.

Asset properties
Configuration > Asset properties

3 selected items. Cancel

Label	Description	Property type
☒ Blade length	Length of the turbine blade in meters	number
☒ Drivetrain type	Mode by which mechanical power is delivered to the generator to produce electricity	enum
☒ Nacelle dimension	Length, width and height of the Nacelle	object
☐ Gearbox specifications	Technical specifications of the gearbox used	object
☐ Tower height	Distance from ground to the middle of turbine's rotor in meter's	number
☐ Platform capacity	Rated or maximum capacity at which rate the turbine produces power in MW	string
☐ Operating data	Operating data	object
☐ Power regulation	Turbine power regulation type	enum
☐ Installation date	Date of installation of turbine	date
☐ Yaw system	Identifier to indicate if Yaw system is present for power regulation	boolean
☐ Model name	Model name of the component	string

1-11 of 11

Cancel Export

TO IMPORT ASSET PROPERTIES

To import asset properties into the DTM application, follow the steps below:

1. On the **Asset properties** page, click **Import**.
2. Upload the JSON file in the drop file area of the resulting dialog. The JSON file is the file containing exported asset properties from a DTM application on another tenant.
3. Click **Next** to preview the asset properties before importing them.
4. Click **Import** to import the asset properties.
5. Alternatively, click **Back** to go to previous step. Hover over the file name and click the **Delete** icon to remove the current file and upload a different one.
6. View newly imported asset properties in the **Asset properties** page.


Import asset properties

Properties 2 of 2 items No filters   Configure columns 

Label	Description	Property type
Electrical	Electrical properties of the wind turbine	object
Sound pressure level	Indicates how loud a wind turbine is.	string

1-2 of 2

Cancel

Back

Import

INFO

When you upload the JSON file, it undergoes validation. If the validation fails, you see a corresponding message in the dialog. Click **Download log file** to view the errors. Review the errors that are reported, resolve them, and attempt the upload again.

You can import a maximum of 500 asset properties at a time.

DEFAULT PROPERTIES

CHARACTERISTICS

- Default properties are readily available when you open the application.
- Default properties cannot be deleted.
- Only selected fields within a default property are editable and the editable fields vary depending on the specific property.
- Default properties cannot be exported.

LOCATION

The **Location** property enables you to assign location (latitude and longitude) to an asset. You can set the default values of the latitude and longitude by entering the corresponding values or alternatively using the map view.

To set the default values for the latitude and longitude using the map view, click **Choose on Map** at the end of the section. Click the full screen icon  at the top right corner of the map to view it in full screen. Without values for latitude and longitude, the marker is hidden. To see the marker, click anywhere on the map. Click or drag the marker to the preferred position to select the default value for latitude and longitude.

INFO

If a property with the label "Location" already exists, the default Location property will be created with the label "Asset Location" and with the key "c8y_Position".

The valid range for the latitude is -90 to +90 and valid range for the longitude is -180 to +180.

The marker on the map is only visible when both values for latitude and longitude are provided.

The default values of both latitude and longitude are automatically updated whenever a new location is selected on the map and vice-versa. The Altitude value is not represented on the map.

ASSET MODELS

The asset model is a blueprint or template to create one or multiple [assets](#). It defines how the asset will look like after it is created. An asset model consists of one or more [asset properties](#) and/or child asset models.

To get a quick start to use the DTM application, import pre-defined asset models from **Asset model samples** page. These asset models and associated properties are defined based on best practices. For details about how to import asset models from samples, see [To import asset model samples](#).

To reach the **Asset models** page, navigate to **Configuration > Asset models**.

When you create an asset model, you see the details of this asset model in the overview.

Model name	Key	Description	Child asset models	Asset properties
Wind turbine AZ-43Y	wind_turbine_az_43y		AZ-43Y Tower, AZ-43Y Nacelle, AZ-43Y Rotor	
AZ-43Y Tower	az_43y_tower			
AZ-43Y Nacelle	az_43y_nacelle		AZ-43Y Low speed shaft, AZ-43Y Gearbox, AZ-43Y Generator, AZ-43Y High speed shaft	Nacelle dimension, Direction type
AZ-43Y Low speed shaft	az_43y_low_speed_shaft			
AZ-43Y Gearbox	az_43y_gearbox			
AZ-43Y Generator	az_43y_generator			
AZ-43Y High speed shaft	az_43y_high_speed_shaft			
AZ-43Y Rotor	az_43y_rotor		AZ-43Y Blade	
AZ-43Y Blade	az_43y_blade			Blade length

In the overview you see the following four categories:

- **Model name:** Name of the asset model
- **Key:** Unique identifier for the asset model
- **Description:** Brief description of the asset model
- **Child asset models:** List of child asset models
- **Asset properties:** List of asset properties

Click **Configure columns** to configure the displayed columns.

Click **Reload** to reload the screen. The total number of asset models is displayed on the bottom left of the page.

Use the **Export asset models** and **Import asset models** feature to transfer asset models between tenants. This is useful for replicating the asset models in multiple tenants. By exporting and importing asset models, you can ensure consistency and reduce the time and effort required for manual configuration. See [To export asset models](#) and [To import asset models](#) on how to export or import asset models.

✓ REQUIREMENTS

ROLES & PERMISSIONS

- To add/update/delete/import asset models: CREATE/ADMIN permission for permission type "Inventory"

DEFAULT ASSET MODELS

Group asset model

Groups are used in Cumulocity to organize devices or other groups. The default group asset model allows any group to be treated as an asset, providing a unified, flexible, and consistent approach to managing assets. Unlike other asset models in the Digital Twin Manager application, the group asset model does not enforce hierarchical relationships.

Characteristics of the group asset model

- Available by default, it cannot be deleted.
- Allows updating of label, description, and icon for the model. This can be restricted by setting the value of the flag **isDefaultGroupEditDisabled** to “true” in the application options. For details, refer to [Application options](#).
- Does not support custom asset properties.
- Since the model doesn’t enforce a hierarchy, it does not require the definition of child asset models.
- The model can be easily identified by the “(Default)” suffix throughout the Digital Twin Manager application.

TO CREATE AN ASSET MODEL

Click **Add asset model** on the top right to create a new asset model. This opens the **New Asset model** page. Here, provide the following six parameters:

Field	Description	Mandatory / Optional
Label	Refers to the name of the asset model. This name is used for the asset model throughout the application.	Mandatory
Key	Used to uniquely identify and store the asset model in the DTM application. Note that the key is automatically generated based on the label. You can modify the automatically generated key to suit to your needs only during the asset model creation. You cannot edit the key after the asset model is created.	Mandatory
Description	Provides a brief description of the asset model.	Optional

Field	Description	Mandatory / Optional
Icon	Displayed next to the label for each asset model. By default, no icon is selected.	Optional
Child asset models	Restricts the asset assignment to the defined models. By default, no subassets are allowed if this field is not set.	Optional
Asset properties	Parameters that define an asset model. By default, no asset properties are selected.	Optional

Once all required asset models are created, you can start creating [assets](#) using these asset models.

To select an icon

1. Click the **Select icon** button on the left.
2. In the following dialog window, search or filter icons by type or select an icon from the predefined list.
3. Select the desired icon.
4. Click **Save**.

To add a child asset model

1. Click **Add child asset model** to add a new child asset model.
2. In the dropdown select one or multiple allowed child asset models.
3. The selected child asset models are then displayed under **Allowed child asset models** with the checkbox **Required** on the right. If the asset model requires this child asset model, select the checkbox. By default, it is clear.
4. Click the remove icon  next to the checkbox to remove the child asset model from the selected asset model.

INFO

If you do not select and check a child asset model, then the asset cannot have any subassets.

If one or more child asset models are checked as required, an asset created with this asset model can have only subassets belonging to the selected child asset models.

Example:

If the asset model is “Rotor” and you select “Blade” as an allowed child asset model, then the newly created asset “AZ-43Y Rotor” can only have “Blade” assets as subassets.

To add an asset property to an asset model

1. Click **Add asset property** to add a new asset property to the asset model.
2. In the resulting dialog box, select one or multiple asset properties from the dropdown field. To search for specific asset properties, type the respective name in the dropdown field.
3. The selected asset properties are then displayed on the **Asset properties** segment with the checkbox **Required** on the right. If the asset model requires this asset property, select the checkbox. By default, it remains clear.
4. Reorder the asset properties via drag and drop.
5. To remove the asset property from the asset model, click the remove icon  next to the checkbox.
6. Click **Save** to save your settings.

INFO

To assign a location to an asset, add the default property Location to the asset model. For more info on default properties, see [Default properties](#).

If there are any validation errors for a field, it is highlighted in red with a validation error message asking you to fill in the required information. Enter the necessary information and click **Save**.

If there is no asset property meeting your requirements when you search, click **New asset property**. For more details on how to create a new asset property see [To create an asset property](#).

ROOT ASSET MODEL

To create a root asset model follow a bottom-up approach by defining all child asset models and asset properties in the provided templates first.

This root asset model can then be used to create an asset hierarchy, which then defines all the assets, subassets, asset properties and devices in the DTM application.

As a result you can use the asset hierarchy in other Cumulocity applications, such as the Cumulocity Cockpit.

TO CREATE AN ASSET MODEL FOR THE ENTIRE HIERARCHY

1. Create all required [asset properties](#).
2. Then [create the asset models](#) in a bottom-up approach, that is, all the child asset models are created first, followed by the root asset model at the end.
3. [Create assets](#) for the new asset hierarchy.

This approach, allows you to create the desired hierarchy including all asset models, child asset models and asset properties in one process.

TO MODIFY AN ASSET MODEL

1. Click the edit icon  on the right side of each asset model.
2. Make your changes in the editor.
3. Click **Save**.
4. If this asset model is used by any asset, select **Confirm** in the resulting confirmation dialog to continue saving your changes.

TO DELETE AN ASSET MODEL

There are two ways to delete an asset model:

Delete each asset model individually

On the **Asset models** page, hover over the desired asset model and click the delete icon  on the right.

Delete the asset models in bulk

On the **Asset models** page, select the checkboxes next to the desired asset models. Click **Delete** to delete the selected asset models.

TO SORT ASSET MODELS

To sort columns, hover over the respective column header and click the sort icon  for an ascending or descending order.

TO FILTER ASSET MODELS

1. Use the filter option to filter asset models based on text in the respective asset group.
2. Hover over a column title for the filter icon  to appear on the right.
3. Click the icon  to filter the asset models in this column.
4. To clear all applied filters click **Clear filters** at the top.

TO EXPORT ASSET MODELS

To export asset models to add in another tenant, follow the steps below:

1. On the **Asset models** page, click **Export**.
2. In the resulting dialog, you see the list of asset models. Initially, all the asset models are collapsed. Click the expand icon  to the left of each asset model to view the next level of child asset models.
3. Select one or more desired asset models.
4. Click **Export** to export the selection.
5. The dialog window closes and a JSON file named 'Export-asset-models.json' downloads.

Use the downloaded file to import the asset models into the DTM application on another tenant.

INFO

You can select a maximum of 30 asset models to export at one time.

When you select an asset model with child asset models and asset properties, all the child asset models and their properties are selected automatically. You cannot unselect a child asset model of an asset model that is already selected. You can export a total of 250 asset models and 500 associated properties at one time.

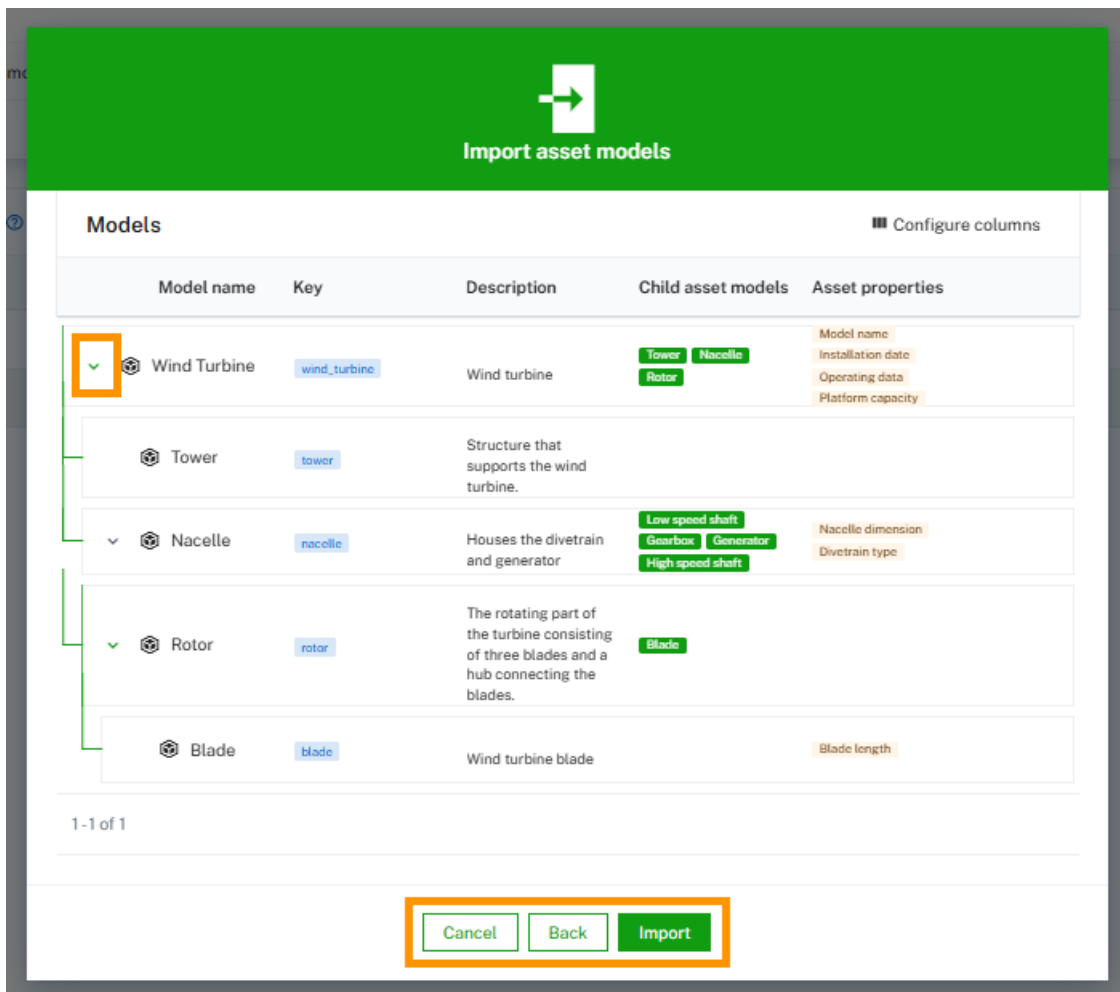
You can select a child asset model to export without selecting its parent. This gets exported as a root asset model.

1 selected item(s).					⌂	✕ Cancel
Model name	Key	Description	Child asset models	Asset properties		
☒ Wind turbine AZ-43Y	wind_turbine_az-43y		AZ-43Y Tower AZ-43Y Nacelle AZ-43Y Rotor			
☒ AZ-43Y Tower	az-43y_tower					
☒ AZ-43Y Nacelle	az-43y_nacelle		AZ-43Y Low speed shaft AZ-43Y Gearbox AZ-43Y Generator AZ-43Y High speed shaft	Nacelle dimension Drivetrain type		
☒ AZ-43Y Rotor	az-43y_rotor		AZ-43Y Blade			
☒ AZ-43Y Blade	az-43y_blade			Blade length		
1-1 of 1						
		Cancel	Export			

TO IMPORT ASSET MODELS

To import asset models, follow the steps below:

1. On the **Asset models** page, click **Import**.
2. Upload the JSON file in the drop file area of the resulting dialog. The JSON file is the file containing exported asset models from a DTM application on another tenant.
3. Click **Next** to preview the asset models. Initially, all the asset models are collapsed. Click the expand icon  to the left of each asset model to view the next level of child asset models.
4. Click **Import** to import the asset models.
5. Alternatively, click **Back** to go to previous step. Hover over the file name and click the remove icon  to remove the current file and upload a different one.
6. The newly imported asset models are listed on the **Asset models** page.



INFO

When you upload the JSON file, it undergoes a validation process. If the validation fails, you see a corresponding message in the resulting dialog box. Click **Download log file** to view the errors. Review the errors that are reported, resolve them, and attempt the import again.

You can import a maximum of 250 asset models and 500 associated properties at one time.

TO IMPORT ASSET MODEL SAMPLES

When you navigate to the **Asset model samples** page following **Configuration > Asset models > Asset model samples**, you see a list of samples with name, description and tags associated with it.

To view the sample, click the menu icon  at the top right of sample name and click **View**. In the resulting dialog, you see a list of asset models. Initially, the asset models in the list are collapsed. Click the expand icon  to the left of each asset model to view the complete hierarchy. Click **Import** to import the asset models in the sample.

Alternatively, to import the asset models in the sample without a preview, click the menu icon  at the top right of sample name and then click **Convert to asset model**.

INFO

When you select **View** or **Convert to asset model**, the asset model sample undergoes a validation process. If the validation fails, you see a corresponding message in the resulting dialog box. Click **Download log file** to view the errors. Review the errors that are reported, resolve them and attempt the action again.



INTRODUCTION

Welcome to the Cumulocity Digital Twin Manager (DTM) application.

This application allows you to create and manage assets around your physical connected devices in Cumulocity using a digital representation of a physical hierarchy of assets and devices. Here, assets can be used to structure and describe devices in logical hierarchies as encountered in the real world environment to help with visualizing the hierarchy in a digital setting.

Once the asset hierarchy is created in the DTM application, the hierarchy can be applied to other applications such as the Cumulocity Cockpit. It saves a lot of time and effort, as the hierarchy is created only once and can be used directly in other Cumulocity hosted applications.

The home screen of the DTM application gives an overview of the application. You can see the number of asset models, asset properties and assets available at a glance. You can add asset properties, asset models and assets using the quick links.

✓ REQUIREMENTS

- Your tenant must be subscribed to the Digital Twin Manager application and the Dtm-ms microservice.

i RELATED TOPICS

- See [Subscribing applications](#) for more information on how to access the DTM application in your tenant.

CONCEPTS

DIGITAL TWIN

A digital twin is a virtual representation of a physical object or system. It is typically used in the context of the Internet of Things (IoT) to help visualize and analyze how a physical object is performing, or to predict how it might behave in the future.

You can use digital twins to simulate and test the performance of complex systems, such as buildings, factories, or even entire cities. By using digital twins, engineers and designers can better understand how a system works and identify potential problems before they occur.

In the Digital Twin Manager application, one asset represents one digital twin.

DIGITAL TWIN ATTRIBUTES

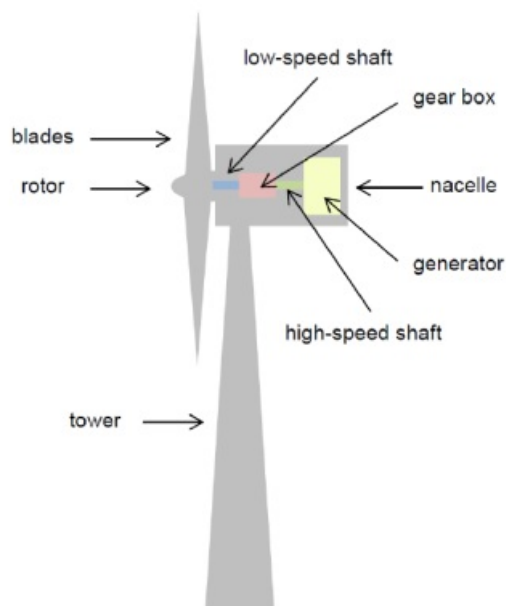
A digital twin contains a number of attributes:

- It represents a unique physical asset.
- It is associated with a single, specific instance of a physical asset.
- It continuously collects data (through sensors, devices, and so on).
- It is continuously connected to the physical asset, updating itself with any change to the asset's state, condition, and context.
- It provides value through visualization, analysis, prediction, and optimization

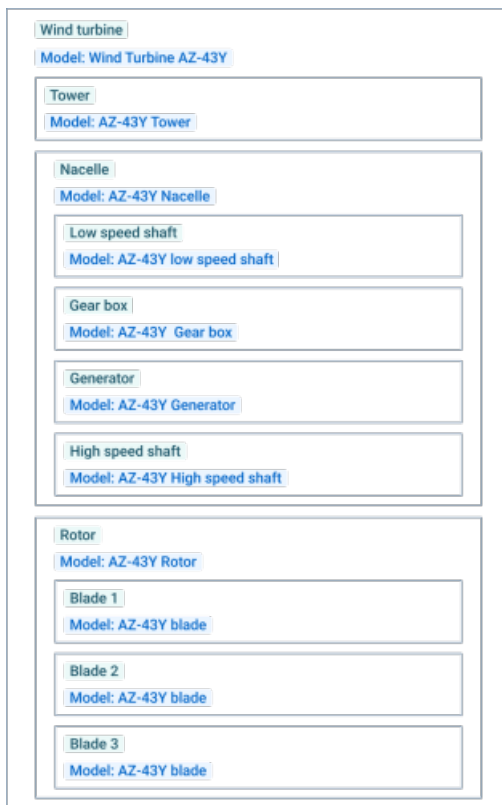
ASSET HIERARCHY

An asset hierarchy is a structure that organizes multiple assets and devices into a hierarchy, allowing for the visualization and management of complex systems.

To create an asset hierarchy, execute the following steps. The example is based on a wind turbine as a physical device.



1. First identify the key systems, processes, and components to be included in order to define the scope of your asset hierarchy.
2. Based on your definitions create the hierarchy of all digital twins.



3. Determine the level of detail needed for each system, process, and component in the hierarchy. This helps to determine the level of accuracy and complexity required for each asset model. For a wind turbine this includes, for example, the tower's height, the length of the blade and so on.
4. To portray the correct structure, first create an asset property for each component in the **Asset properties**, see [To create an asset property](#) for details. The asset properties are later used in creating the individual asset models.
5. To create an asset model work from the smallest component to the largest. For a wind turbine that means to start with the asset model "blade", followed by "rotor", "tower", and "nacelle" and finally "wind turbine". Assign the required asset property to each asset model as well as the allowed contained asset models (for example, a wind turbine requires a rotor, which in turn requires rotor blades).
6. Afterwards, add the necessary asset instances. This process starts from the outer most level (for example, the wind turbine) and ends at the deepest level (for example, the blades).
7. Test and validate the asset hierarchy to ensure that it accurately represents the real-world system or process. This can involve simulating different scenarios and comparing the results to real-world data.
8. Continuously update and maintain the asset hierarchy as changes occur in the real-world system or process. This ensures that the hierarchy remains accurate and up to date.